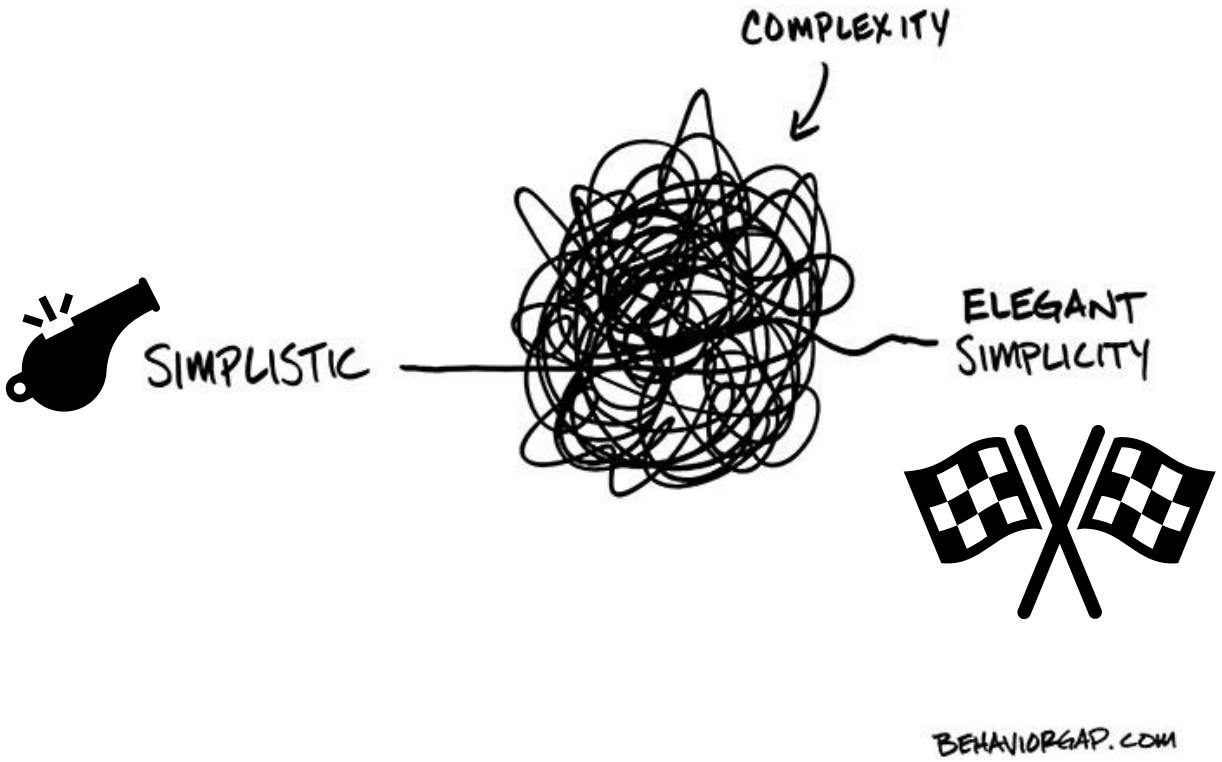
PT4052 | PT7004

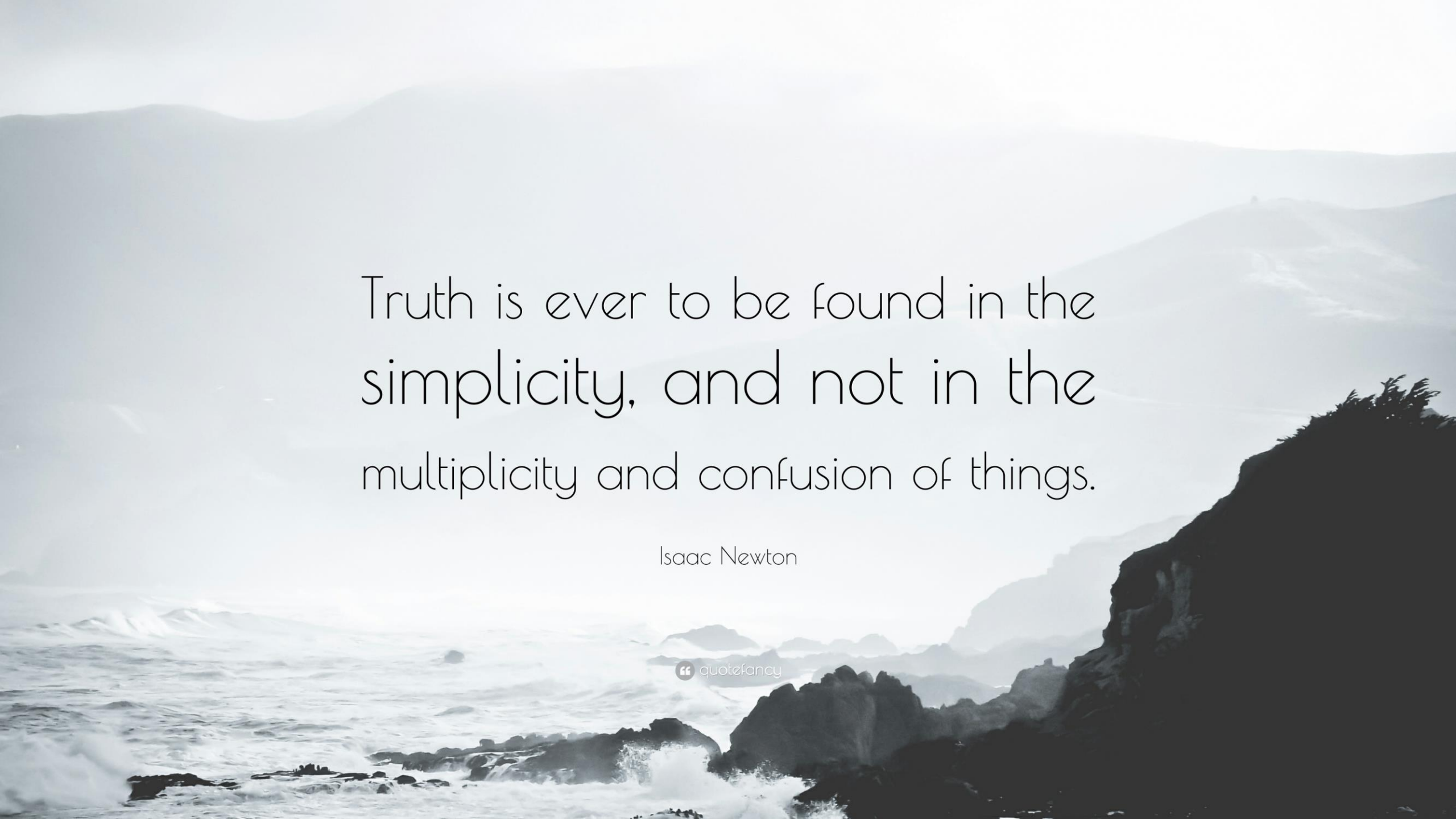
Principles of Assessment in Acute Muscle Injuries

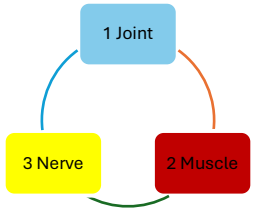
Management of Acute Muscle injuries

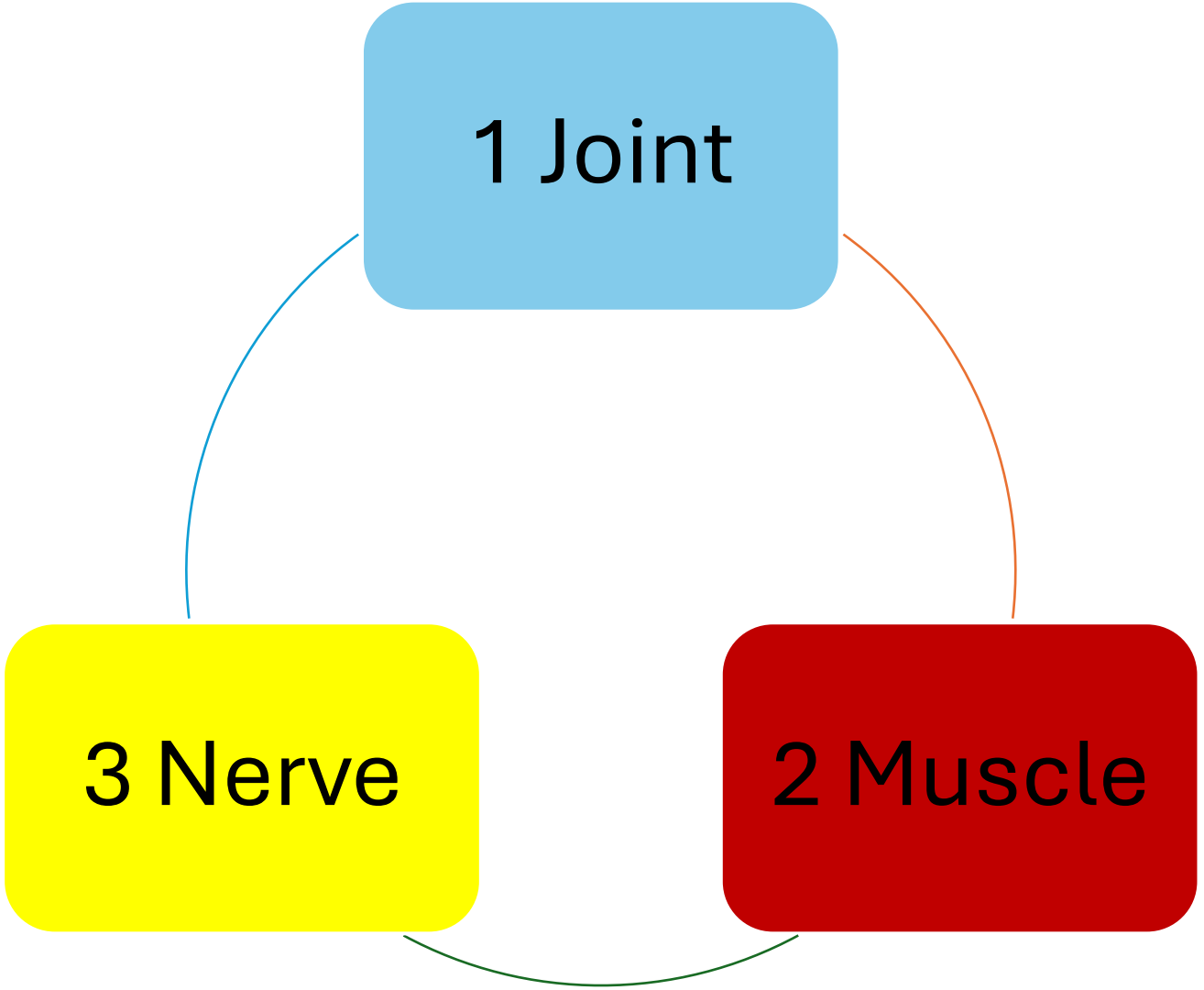
Explain	The important of the subjective and objective components of assessment for acute muscle injuries
Describe	The how pain levels and mechanism of injury might help to guide a tentative diagnosis and the type of injury.

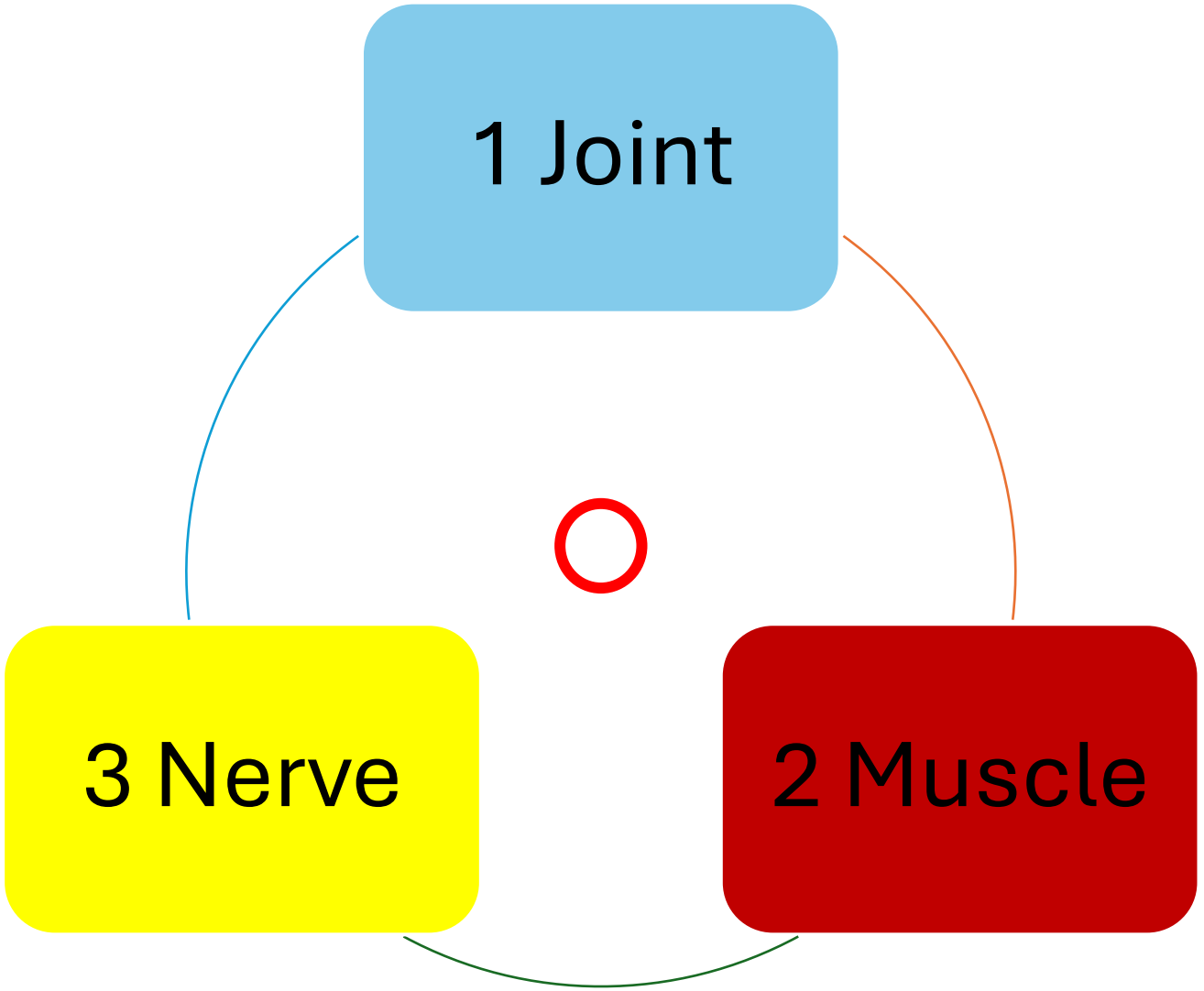


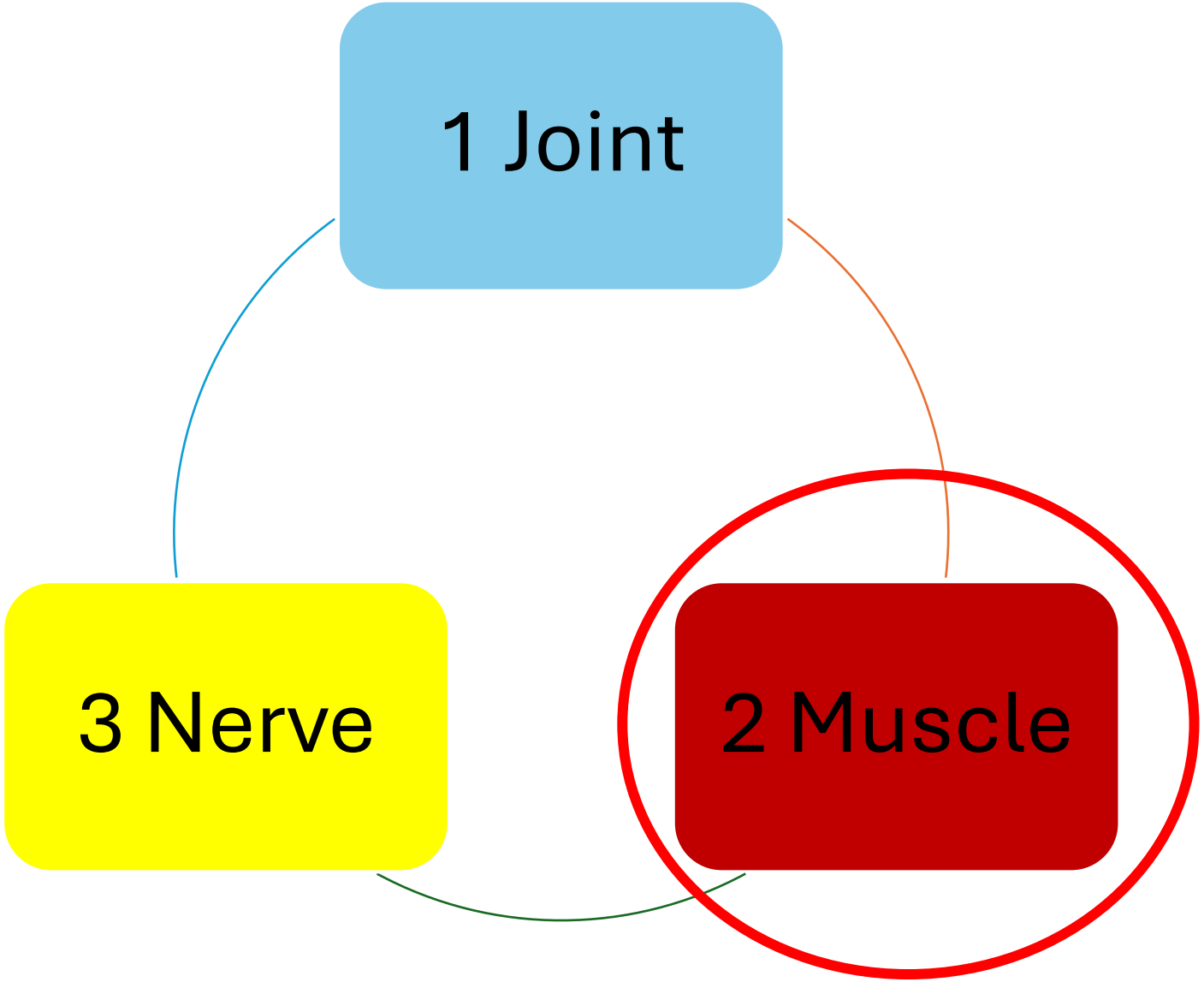
Musculoskeletal Assessment


Truth is ever to be found in the
simplicity, and not in the
multiplicity and confusion of things.
Isaac Newton
quotefancy


1 Joint
2 Muscle
3 Nerve


1 Joint
2 Muscle
3 Nerve


1 Joint
3 Nerve
2 Muscle


1 Joint
3 Nerve
2 Muscle

S O A P

S O A P: Subjective

Why do we need to take a history?

- To inform diagnostic reasoning
- To inform prognosis
- To identify risk (i.e. sinister pathology or risk of poor outcome).
- To identify contributing factors (e.g. activity-related or psychosocial)
- To inform the “objective” physical exam.
- To inform management plan.

- Developing a therapeutic relationship – it can be a treatment itself!

- If the patient has referral information, check that this is correct – e.g. patient has been referred for condition X.
  - The patient's situation may have changed

- PC: If there is no referral information, establish the Presenting Condition (PC:)

- HPC: History of Presenting Condition
  - Why is this important?
  - Timelines, reoccurrence

- Mechanism of injury (traumatic or atraumatic/insidious).
  - Why is this important?
  - Muscles involved, biaxial

- Diagnostic Imaging: X-ray/MRI/Ultrasound etc.
  - Why is this important if it is available?
  - Cross Sectional Area of Damage (CSA).

	BAMIC
Gradel 0 (a/b)	Imaging normal/patchy edema
Grade 1a (myofascial), 1b (myotendinous)	<10% CSA; <5 cm of length
Grade 2a (myofascial), 2b (myotendinous)	10%-50% CSA; 5-15 cm of length; fiber disruption 1-5 cm
Grade 2c (tendinous)	<50% tendinous CSA; <5 cm of length
Grade 3a (myofascial), 3b (myotendinous)	>50% CSA; >15 cm of length; fiber disruption >5 cm
Grade 3c (tendinous)	>50% tendinous CSA; <5 cm of length
Grade 4c	Complete tear

S O A P

S I N, Severity, Irritability, Nature

Severity (for all symptoms):

- Pain @ worst out of ten (Visual Analogue Scale)
- Pain @ best out of ten (Visual Analogue Scale) (if the pain is constant and unremitting – this would be a concern – e.g. cancer)

Muscle Injuries

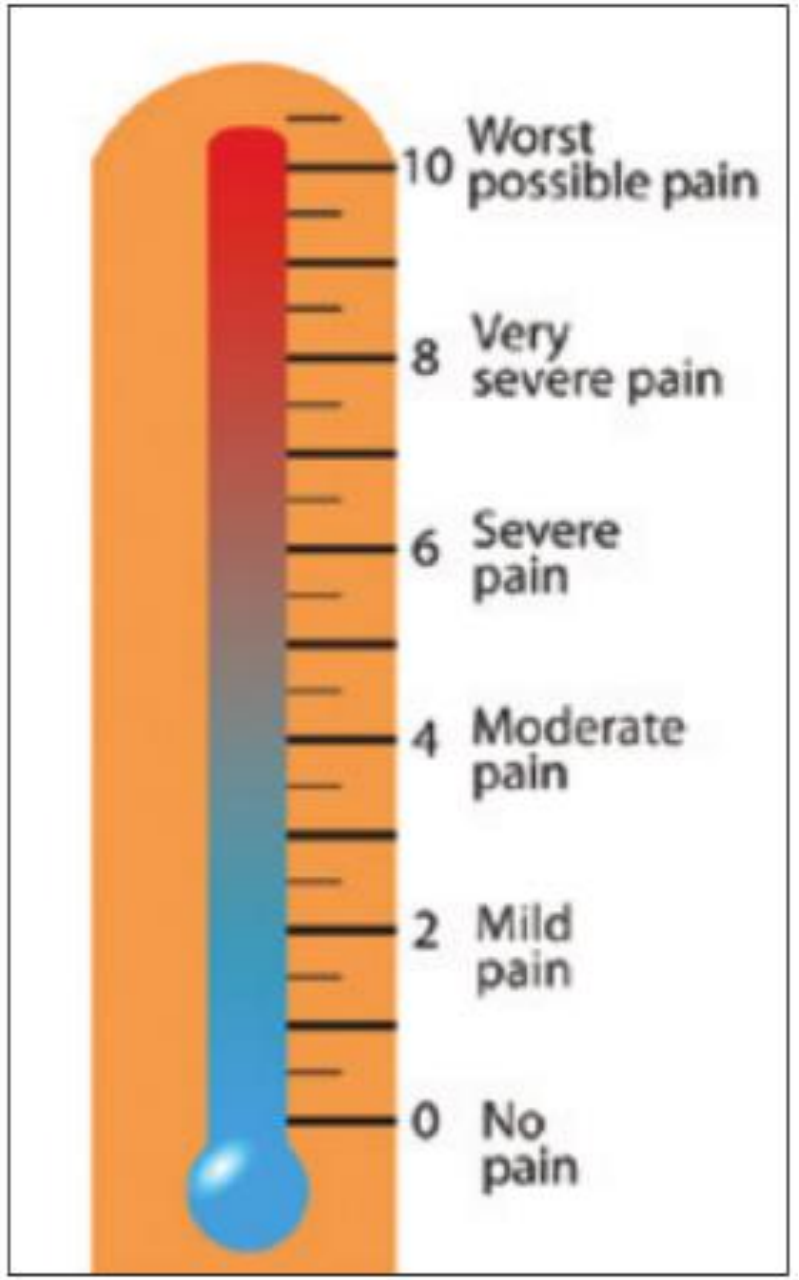
Grade I

Grade II

Grade III



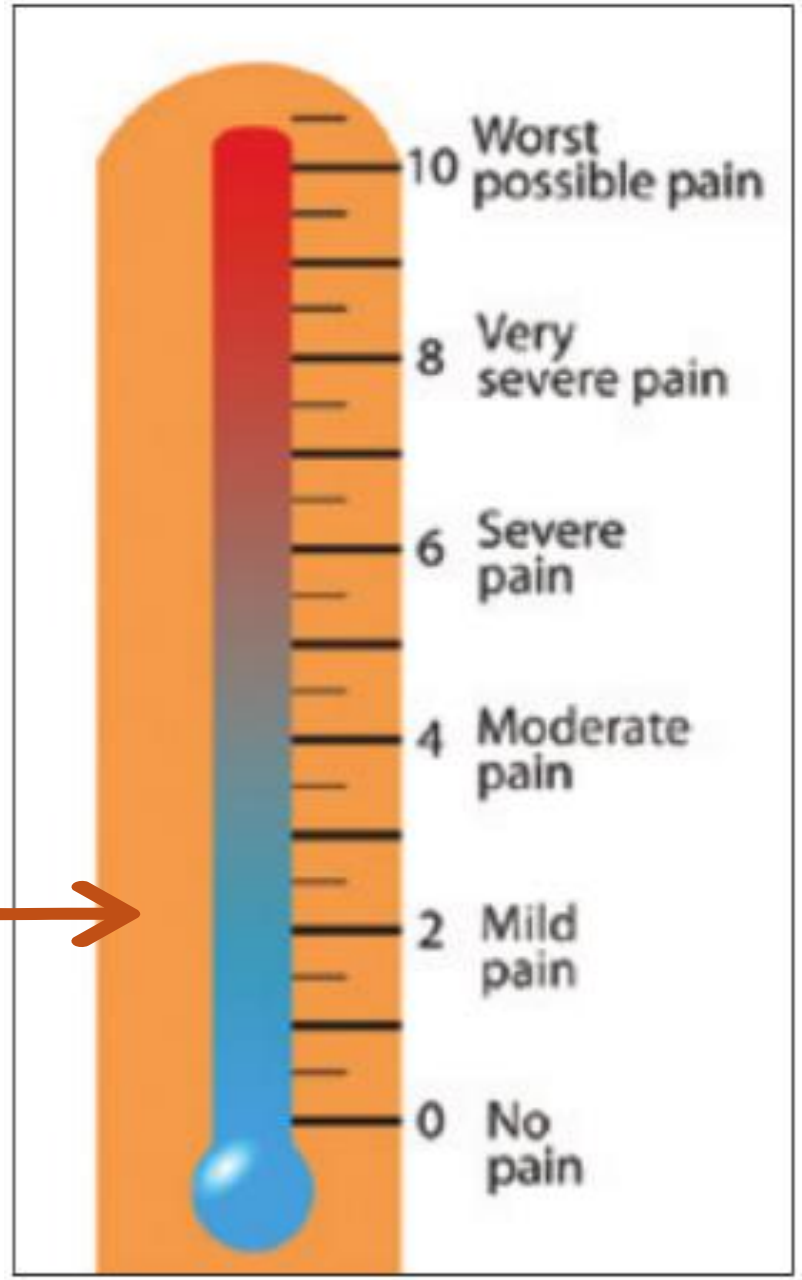
How the subjective might guide your subsequent assessment:



Grade I

Discomfort

Oedema



How the subjective might guide your subsequent assessment:

Grade II

Loss of Function

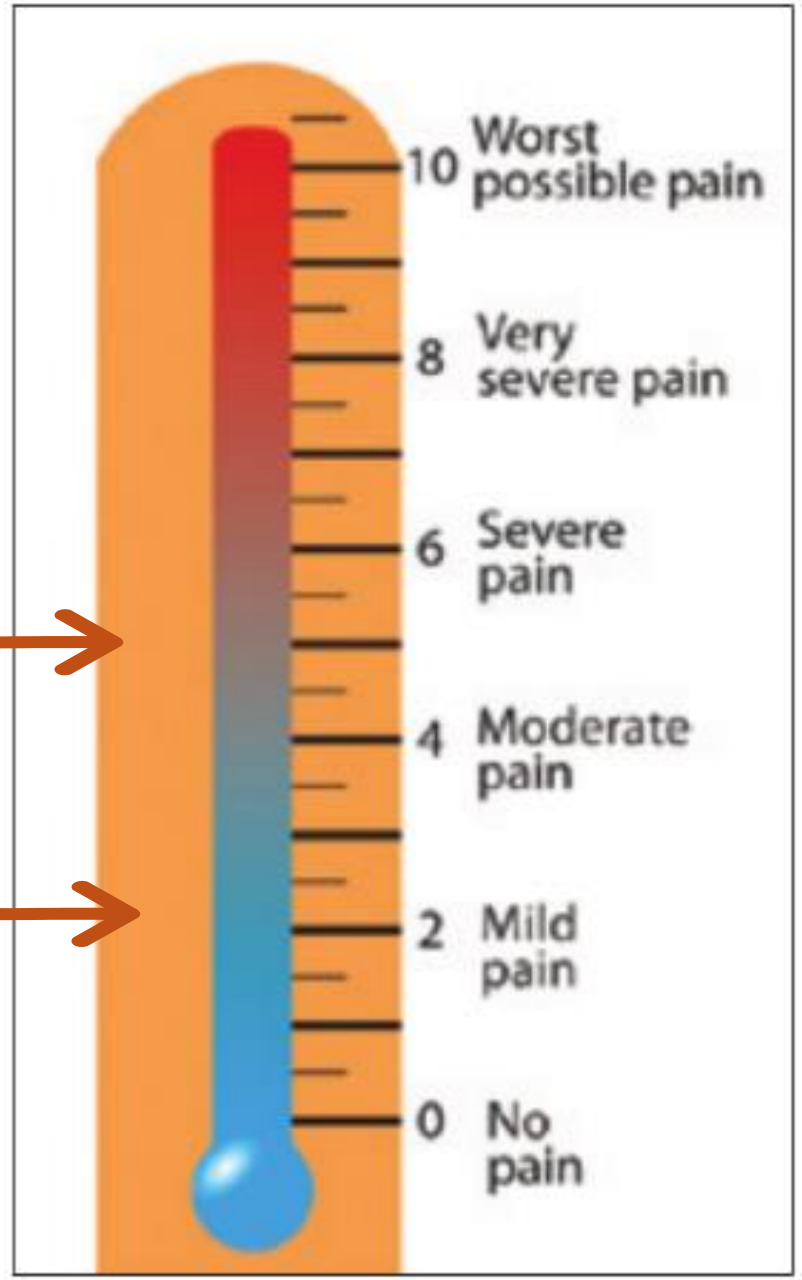
Gaps (may be palpable)

Ecchymosis possible

Grade I

Discomfort

Oedema



How the subjective might guide your subsequent assessment:

Grade III

Severe Pain

Extensive Haematoma

Complete Rupture

Grade II

Loss of Function

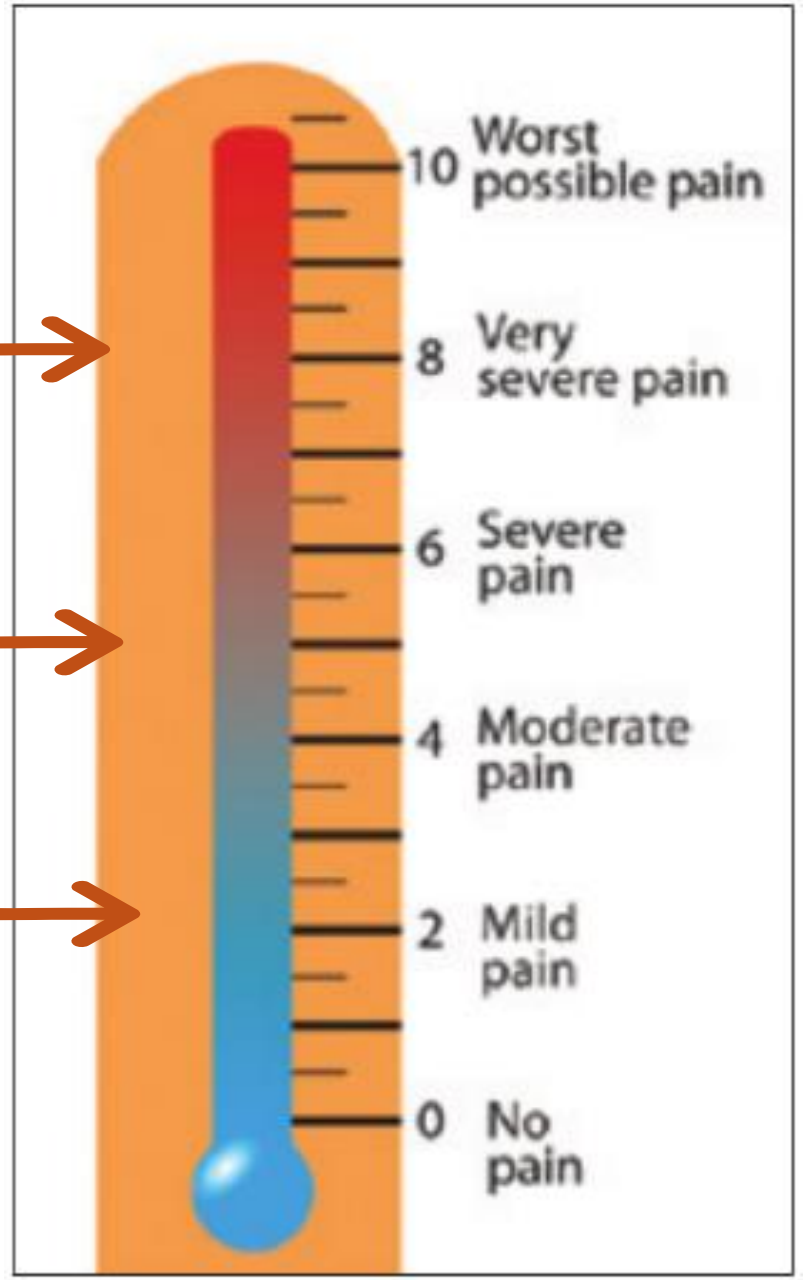
Gaps (may be palpable)

Ecchymosis possible

Grade I

Discomfort

Oedema



How the subjective might guide your subsequent assessment:

S O A P

Irritability:

- When the pain comes and it takes more than a few minutes to go away, the pain is irritable – we would be more selective in the tests we might use.

Nature: Quality of pain

- Sharp?
- Shooting?
- Pins & Needles?
- Numbness?
- Dull Ache?
- Deep

S O A P

Aggs & Easing?

Aggravating: What activities make it worse?
Prompts: Standing / Walking / Loading

Easing: What activities make it better?
Prompts: Rest? (Movement doesn’t symptoms)

S O A P

24 hour pattern:

AM – PM
Intermittent – activity dependent
AM – stiff/worse - inflammation
Aggravated by movement – mechanical/or inflammatory worsened by movement.
Constant?? - Pain unaffected by rest or activity – systemic (bone pain).

S O A P

- **Observation:**
  AROM: Active range of movement
- **PROM: Passive range of movement**
- **Joint integrity tests:**
- **Muscle tests:**
- **Nerve tests:**
- **Special tests:**
- **Palpation:**
- **Joint Tests:**

S O A P

- Observation:
  AROM: Active range of movement
- ~~PROM: Passive range of movement~~
- ~~Joint integrity tests:~~
- Muscle tests: Isometric?
- ~~Nerve tests:~~
- ~~Special tests:~~
- Palpation:
- ~~Joint Tests:~~

S O A P

Observation:

- **Sitting:**
  - **frontal plane (symmetry) – level of shoulders, hips, sacral sitting**
  - **Sagittal – kyphosis, lordosis, loss of lumbar curve**
- **Standing**
  - **Frontal**
    - **Head position, Level of shoulders, Level of hips**
    - **Weight-bearing, feet, ankles relative to heel, Foot pronation?**
    - **Knee valgus/varus**
    - **Scoliosis (posterior)**
  - **Sagittal**
    - **Kyphosis**
    - **Lordosis**
    - **Pelvic Tilt**
    - **Knees (neutral, locked, hyperextended)**

S O A P

Observation:

- **Sitting:**
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    - **Knees (neutral, locked, hyperextended)**

Arjen Robben





Season	Injury	from	until	Days	Games missed
04/05	Knock	Aug 10, 2004	Oct 18, 2004	69 days	16
04/05	Knock	Feb 3, 2005	Mar 16, 2005	41 days	9
07/08	Muscle tear	Oct 18, 2007	Nov 1, 2007	14 days	4
07/08	Ankle injury	Mar 3, 2008	Mar 24, 2008	21 days	4
08/09	Achilles tendon irritation	Oct 6, 2008	Oct 20, 2008	14 days	3
08/09	Torn muscle fiber	Nov 5, 2008	Nov 27, 2008	22 days	7
09/10	Knee problems	Oct 1, 2009	Oct 22, 2009	21 days	4
09/10	Knee problems	Nov 12, 2009	Nov 30, 2009	18 days	5
09/10	strain	Mar 29, 2010	Apr 1, 2010	3 days	1
09/10	Thigh problems	Jun 7, 2010	Jun 21, 2010	14 days	2
10/11	Muscle tear	Aug 5, 2010	Jan 3, 2011	151 days	33
10/11	cold	Feb 3, 2011	Feb 7, 2011	4 days	1
11/12	Capsular injury	Jul 28, 2011	Aug 4, 2011	7 days	1
11/12	Back problems	Aug 11, 2011	Aug 15, 2011	4 days	1
11/12	Inflammation of pubic bone	Aug 22, 2011	Sep 22, 2011	31 days	7

https://www.transfermarkt.co.uk/arjen-robben/verletzungen/spieler/4360/plus/1

Season	Injury	from	until	Days	Games missed
11/12	Inguinal hernia	Oct 6, 2011	Nov 14, 2011	39 days	10
11/12	influenza	Dec 5, 2011	Dec 8, 2011	3 days	1
12/13	cold	Aug 30, 2012	Sep 3, 2012	4 days	1
12/13	Groin problems	Sep 10, 2012	Sep 13, 2012	3 days	1
12/13	Thigh problems	Sep 13, 2012	Sep 17, 2012	4 days	1
12/13	muscular problems	Sep 27, 2012	Oct 25, 2012	28 days	9
12/13	Torn muscle fiber	Nov 15, 2012	Dec 3, 2012	18 days	5
12/13	Back problems	Dec 6, 2012	Dec 31, 2012	25 days	3
12/13	Calf stiffness	Mar 4, 2013	Mar 11, 2013	7 days	1
12/13	muscular problems	May 2, 2013	May 6, 2013	4 days	1
13/14	Capsular injury	Sep 2, 2013	Sep 5, 2013	3 days	-
13/14	Groin problems	Oct 28, 2013	Nov 7, 2013	10 days	2
13/14	flesh wound	Dec 5, 2013	Jan 16, 2014	42 days	5
13/14	muscular problems	Jan 27, 2014	Jan 30, 2014	3 days	1
14/15	Tendonitis	Aug 25, 2014	Sep 15, 2014	21 days	5

https://www.transfermarkt.co.uk/arjen-robben/verletzungen/spieler/4360/plus/1

Season	Injury	from	until	Days	Games missed
14/15	Knee problems	Sep 18, 2014	Sep 22, 2014	4 days	1
14/15	muscular problems	Sep 18, 2014	Sep 22, 2014	4 days	1
14/15	Thigh problems	Oct 23, 2014	Oct 27, 2014	4 days	1
14/15	pinched nerve	Mar 12, 2015	Mar 19, 2015	7 days	1
14/15	Injury to abdominal muscles	Mar 23, 2015	Apr 27, 2015	35 days	9
14/15	Torn muscle bundle	Apr 30, 2015	Jun 29, 2015	60 days	8
15/16	Adductor injury	Sep 3, 2015	Oct 22, 2015	49 days	17
15/16	muscular problems	Nov 26, 2015	Dec 21, 2015	25 days	6
15/16	influenza	Jan 7, 2016	Jan 11, 2016	4 days	-
15/16	minor knock	Feb 4, 2016	Feb 4, 2016	0 days	-
15/16	Rest	Mar 10, 2016	Mar 14, 2016	4 days	1
15/16	Adductor pain	Mar 14, 2016	Jun 6, 2016	84 days	22
16/17	Torn muscle fiber in the adductor area	Jul 18, 2016	Sep 12, 2016	56 days	8
16/17	Thigh problems	Nov 14, 2016	Nov 24, 2016	10 days	2
17/18	Calf strain	Jul 6, 2017	Aug 10, 2017	35 days	1

https://www.transfermarkt.co.uk/arjen-robben/verletzungen/spieler/4360/plus/1

Season	Injury	from	until	Days	Games missed
17/18	influenza	Sep 18, 2017	Sep 21, 2017	3 days	1
17/18	Torn muscle fiber	Nov 23, 2017	Dec 11, 2017	18 days	4
17/18	Sciatica problems	Dec 4, 2017	Dec 18, 2017	14 days	4
17/18	Rest	Dec 18, 2017	Dec 21, 2017	3 days	1
17/18	pinched nerve	Mar 12, 2018	Mar 19, 2018	7 days	2
17/18	Adductor injury	Apr 26, 2018	Jul 3, 2018	68 days	5
18/19	Back problems	Oct 28, 2018	Nov 2, 2018	5 days	1
18/19	Knee problems	Nov 5, 2018	Nov 21, 2018	16 days	4
18/19	Thigh problems	Nov 30, 2018	Feb 3, 2019	65 days	9
18/19	Fitness	Feb 4, 2019	Mar 4, 2019	28 days	6
18/19	Calf injury	Mar 5, 2019	Apr 25, 2019	51 days	11
20/21	Groin injury	Sep 14, 2020	Oct 16, 2020	32 days	3
20/21	Calf injury	Oct 17, 2020	Apr 4, 2021	169 days	25

https://www.transfermarkt.co.uk/arjen-robben/verletzungen/spieler/4360/plus/1

Netherlands 5-1 Spain (Jun 13, 2014) https://www.youtube.com/watch?v=2KrHsFgmErk

SB SPORTS BRIEF

16. Cristiano Ronaldo - 33.95 km/h
15. Antonio Valencia - 35.1 km/h
14. Leroy Sané - 35.9 km/h
13. Kylian Mbappé - 36 km/h
12. Ronaldo Nazario - 36.2 km/h
11. Mohamed Salah – 36.64 Km/h
10. Antonio Rüdiger - 36.72 km/h
9. Dara O'Shea - 36.73km/h
8. Dominik Szoboszlai - 36.76km/h
7. Pedro Neto - 36.86km/h
6. Gareth Bale - 36.9 km/h
5. Chiedozie Ogbene - 36.93km/h
4. Arjen Robben - 37 km/h
3. Kyle Walker - 37.31km/h
2. Micky van de Ven - 37.38km/h
1. Thierry Henry - 39.2km/h

https://sportsbrief.com/football/41991-the-top-fastest-soccer-players-unveiling-speedsters/

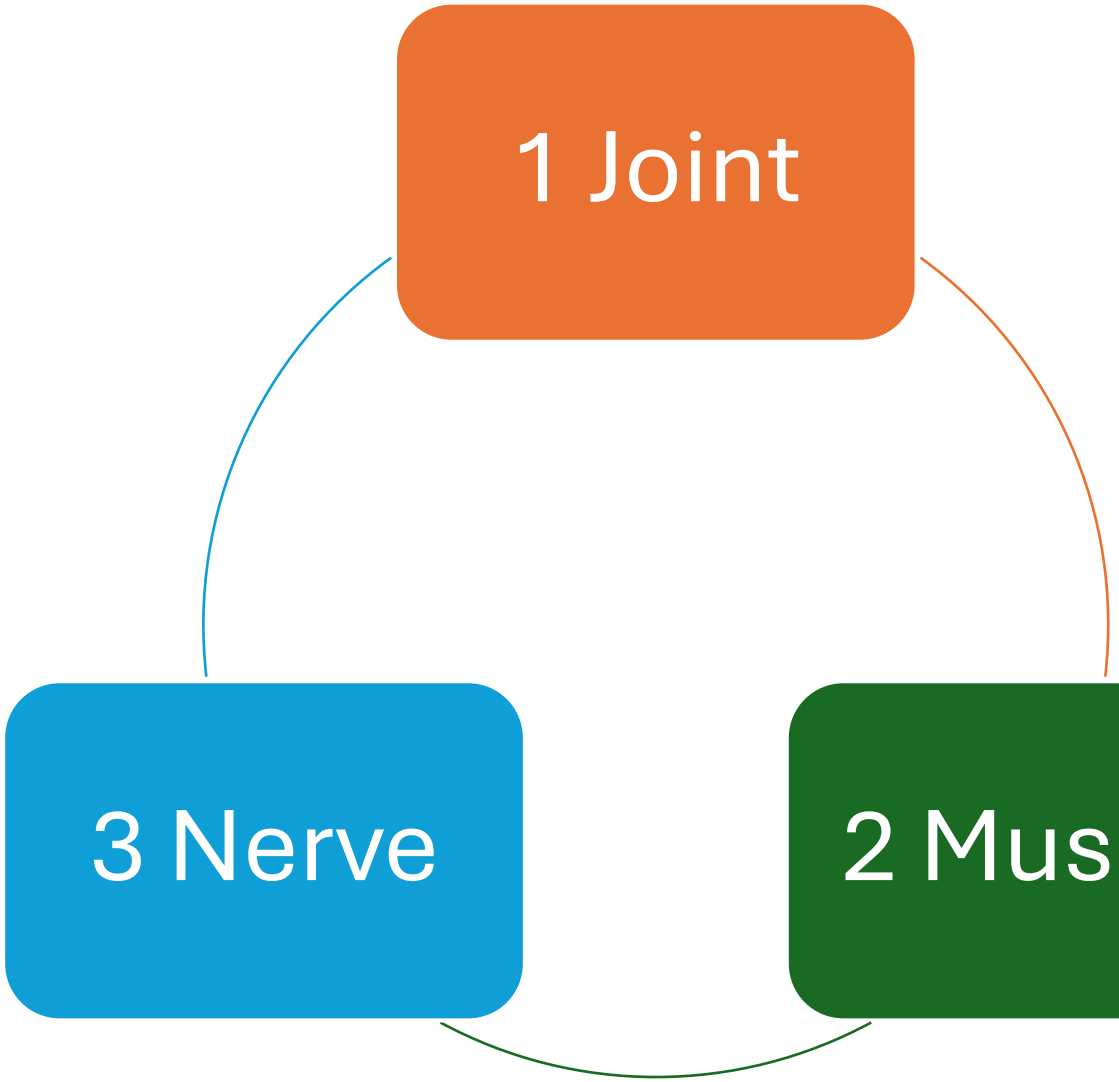
S O A P

Palpation: The clinician must be aware of the psychological impact of touch for a patient. Palpation will have a neurophysiological effect and this may produce a change in the patient's symptoms. During the palpation of soft tissues and skeletal tissues, the following should be noted:

- **the temperature of the area (increase is indicative of local inflammation)**
- **the presence of oedema and effusion**
- **‘gap’ in the muscle**
- **mobility and feel of superficial tissues**
- **the presence or elicitation of muscle spasm**
- **tenderness of bone, ligament, muscle, tendon, tendon sheath, trigger point and nerve**
- **increased or decreased prominence of bones**
- **joint effusion or swelling of a limb can be measured with tape measures, comparing left & right sides**
- **pain provoked or reduced on palpation.**

Diagnostic use of palpation has been questioned due to its subjectivity with issues of reliability as well as validity. However when combined with clinical reasoning it can yield important information. Palpation findings can be recorded on a body chart.

Mechanical	Specific movements/actions aggravating or easing the pain Intensity of stimulus and pain are related Usually pain is localized (small receptive field)
Inflammatory	May be associated with swelling redness and heat Pain may be present at rest Hyperalgesia/allodynia as a response to stimulus Diffuse pain areas/not localized (large receptive field) Close association with acute pain and tissue damage Good response to antiinflammatory medication
Ischaemic	Pain associated with prolonged and unusual postures Rapid reduction of symptoms with change of position Worsening of symptoms at the end of the day or after accumulation of activity Usually no evidence of acute tissue injury Poor response to antiinflammatory medication



- **Muscles: Observation, Active range of movement, muscle tests, palpation**

Signs & Symptoms

Observation	Swelling, bruising, discolouration
Quick Touch	Heat
Range of movement	Reduced ROM, painful
Manual Muscle Tests	Reduced strength, painful
Palpation	Tautness, pain, gaping

S O A P

Muscle			
	Active movement	Reproduces patient's symptoms	Reduced strength
		Poor quality	
Passive physiological movements	Patient's symptoms not reproduced		
Isometric contraction	Reproduces patient's symptoms	Reduced strength	
		Poor quality	
Passive lengthening of muscle	Reproduces patient's symptoms	Reduced range Increased resistance Decreased resistance	
Palpation of muscle	Reproduces patient's symptoms	Tenderness	
Reassessment following therapeutic dose of	Improvement in tests which	No change in physical tests which reproduce	

Graded I, II, III depending on certain factors

Injury – Muscle Strains

Grade I	Grade II	Grade III
<25% fibres torn	25-75% fibres torn	Complete tear
Localised pain	Severe pain and swelling	No pain?
No/minimal loss of strength	Strength reduced	Loss of muscle function – 0/5
	ROM reduced	Often at MTJ – Avulsion?
	Pain on muscle contraction	Possible deformity

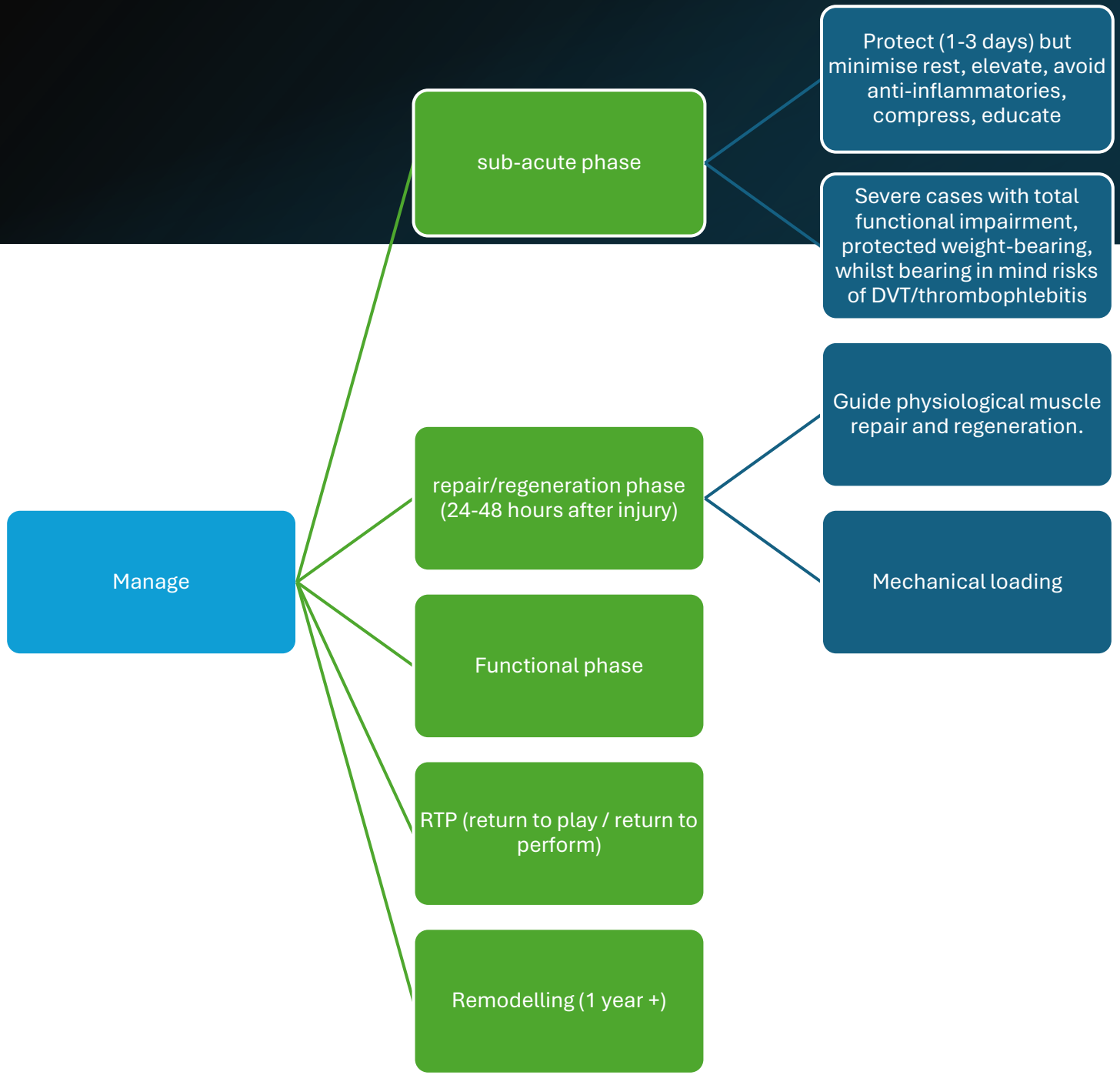
Injury – Others

Compartment syndromes

- Overuse
- A sheath around a collection of muscles
- Typically anterior lower leg
- Pressure increases inside the sheath

Physical exam

- should focus on the neurovascular territory of the involved compartment:
- Observe skin for lesions, swelling, or colour change
- Palpate over the compartment, observing temperature, tension, tenderness
- Check pulses
- Evaluate two-point discrimination and sensation
- Evaluate motor function


Manage
sub-acute phase
Protect (1-3 days) but minimise rest, elevate, avoid anti-inflammatories, compress, educate
Severe cases with total functional impairment, protected weight-bearing, whilst bearing in mind risks of DVT/thrombophlebitis
repair/regeneration phase (24-48 hours after injury)
Guide physiological muscle repair and regeneration.
Mechanical loading
Functional phase
RTP (return to play / return to perform)
Remodelling (1 year +)

Case Study Management

From the available information in the case study discuss:

- The grade of injury (I, II, or III)
- BAMIC grade (0, 1a/b, 2a/b/c, 3 a/b/c, 4c)
- The timeframe of the injury, and ***rehabilitation focus***
  - Haematoma (hours)
  - 1-3 days inflammation ***(protect)***
  - 24-48 hours+ post injury ***(optimal loading)***
  - Romodelling (upto 1+year) ***(functional/RTP)***
- Although it is early in the course, what could a treatment plan look like for the case study?

References

- Bao W, Sun Y, Zhang T, Zou L, Wu X, Wang D, Chen Z. Exercise Programs for Muscle Mass, Muscle Strength and Physical Performance in Older Adults with Sarcopenia: A Systematic Review and Meta-Analysis. Aging Dis. 2020 Jul 23;11(4):863-873. doi: 10.14336/AD.2019.1012. PMID: 32765951; PMCID: PMC7390512.
- Brukner, P. (2012). Brukner & Khan's clinical sports medicine. North Ryde: McGraw-Hill.
- Danielsson, A., Horvath, A., Senorski, C., Alentorn-Geli, E., Garrett, W. E., Cugat, R., ... & Hamrin Senorski, E. (2020). The mechanism of hamstring injuries–a systematic review. BMC musculoskeletal disorders, 21, 1-21.
- Edouard, P., Reurink, G., Mackey, A. L., Lieber, R. L., Pizzari, T., Järvinen, T. A., ... & Hollander, K. (2023). Traumatic muscle injury. Nature Reviews Disease Primers, 9(1), 56.
- Green, B., & Pizzari, T. (2017). Calf muscle strain injuries in sport: a systematic review of risk factors for injury. British journal of sports medicine, 51(16), 1189-1194.
- Haws, B. E., Luo, T. D., Al'Khafaji, I. M., Rogers, J. P., Botros, D. B., & Freehill, M. T. (2017). Definitive management of thigh contusions in athletes: but how definitive? A systematic review. Journal of ISAKOS, 2(2), 67-74.
- Järvinen, T. A., Järvinen, M., & Kalimo, H. (2013). Regeneration of injured skeletal muscle after the injury. Muscles, ligaments and tendons journal, 3(4), 337.
- Mechó, Sandra & Balius, Ramon & Bossy Langella, Mireia & Valle, Xavier & Pedret, Carles & Ruiz-Cotorro, Ángel & Rodas, Gil. (2023). Isolated Adductor Magnus Injuries in Athletes: A Case Series. Orthopaedic Journal of Sports Medicine. 11. 232596712211388. 10.1177/23259671221138806.
- Petty, N. J., & Barnard, K. (Eds.). (2017). Principles of Musculoskeletal Treatment and Management E-Book: Principles of Musculoskeletal Treatment and Management E-Book. Elsevier Health Sciences.
- Pollock, N., James, S. L., Lee, J. C., & Chakraverty, R. (2014). British athletics muscle injury classification: a new grading system. British journal of sports medicine, 48(18), 1347-1351.